

Exp no-4.b

Random Forest

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[24]
✓ 4s
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score,
    f1_score, confusion_matrix, classification_report,
    roc_auc_score, roc_curve
)
import pandas as pd
import matplotlib.pyplot as plt
data = load_breast_cancer()
X = pd.DataFrame(data.data, columns=data.feature_names)
y = data.target
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.20,
    random_state=42,
    stratify=y
)
rf = RandomForestClassifier(random_state=42)
parameters = {
    'n_estimators': [100],
    'max_depth': [None, 10],
    'min_samples_split': [2],
    'min_samples_leaf': [1]
}
grid = GridSearchCV(
    rf,
    parameters,
    scoring='f1',
    cv=3
)
grid.fit(X_train, y_train)
print("Best Parameters:", grid.best_params_)
model = grid.best_estimator_
y_pred = model.predict(X_test)
y_prob = model.predict_proba(X_test)[:, 1]
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
auc = roc_auc_score(y_test, y_prob)
cm = confusion_matrix(y_test, y_pred)
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print( "\n==== RANDOM FOREST RESULTS ===== )
print("Accuracy :", round(accuracy, 4))
print("Precision:", round(precision, 4))
print("Recall   :", round(recall, 4))
print("F1-Score :", round(f1, 4))
print("ROC-AUC  :", round(auc, 4))
print("\nConfusion Matrix:")
print(cm)
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
importance = pd.Series(
    model.feature_importances_,
    index=X.columns
)
top10 = importance.sort_values(
    ascending=False
).head(10)
plt.figure(figsize=(8, 5))
top10.sort_values().plot(kind='barh')
plt.xlabel("Importance")
plt.title("Top 10 Feature Importances")
plt.show()
fpr, tpr, threshold = roc_curve(y_test, y_prob)
plt.figure()
plt.plot(
    fpr,
    tpr,
    label="Random Forest AUC = " + str(round(auc, 3))
)
plt.plot([0, 1], [0, 1], '--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve - Random Forest")
plt.legend()
plt.grid()
plt.show()

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... Best Parameters: {'max_depth': None, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 100}

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===== RANDOM FOREST RESULTS =====

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Accuracy : 0.9561
Precision: 0.9589
Recall   : 0.9722
F1-Score : 0.9655
ROC-AUC  : 0.9937

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Confusion Matrix:

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[[39  3]
 [ 2 70]]

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Classification Report:

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	precision	recall	f1-score	support
0	0.95	0.93	0.94	42
1	0.96	0.97	0.97	72
accuracy			0.96	114
macro avg	0.96	0.95	0.95	114
weighted avg	0.96	0.96	0.96	114

